PM599: AI as a Tool for Biostatistics

Week 1 — Introduction to LLMs and Generative AI

Lu Zhang · Division of Biostatistics · USC Keck School of Medicine

May 21, 2026 (Thursday)

Welcome to PM599

What this course is about:

Using state-of-the-art AI tools to assist biostatisticians and data scientists — in code writing, data analysis, scientific communication, and critical thinking.

What this course is not:

- A machine learning or deep learning course
- A course on building AI systems
- A replacement for statistical rigor or domain expertise

> The goal is to make you a **thoughtful, effective, and responsible** user of AI tools in biostatistical research.

Six Weeks, Six Topics

Week	Dates	Topic
1	Thu 5/21, Tue 5/26	**Intro to LLMs · Generative AI in Biostatistics**
2	Thu 5/28, Tue 6/2	Prompt Engineering
3	Thu 6/4, Tue 6/9	AI-Assisted Data Cleaning · Data Privacy
4	Thu 6/11, Tue 6/16	Statistical Analysis · Bias
5	Thu 6/18, Tue 6/23	Figures and Tables
6	Thu 6/25, Tue 6/30	Critiquing AI Summaries · Ethics · Presentations

Lectures: Thursdays · Labs: Tuesdays · 1:00–2:50 PM · SSB-114

Grading

Component	Weight	Details
Homework (×4)	40%	One per topic area; individual work
Class Participation	30%	Attendance, in-class activities, discussion
Final Project	30%	AI-assisted analysis + 5-minute video presentation

Grading scale: A ≥ 93 · A– 90–92 · B+ 87–89 · B 83–86 · B– 80–82 · C+ 77–79 · C 73–76 · F < 60

Required materials: Laptop · R (r-project.org) · Brightspace

Homework and Final Project

Homework (4 assignments, 10 pts each):

- HW1: Compare AI tools and critique output *(due May 28)*
- HW2: Simulation study evaluating statistical theory *(Week 2)*
- HW3: Data cleaning exercise *(Week 3)*
- HW4: Statistical analysis with AI assistance *(Week 4)*

Final Project (30%):

- AI-assisted analysis on a topic of your choice
- 5-minute recorded video presentation; must include AI disclosure statement
- Due last week of class (June 25–30)

Academic Integrity

> You **may** use AI tools for assignments in this course.
> You **must disclose** how, following the professional standards we discuss today.
> Undisclosed AI use violates the USC Student Handbook.

This course holds you to the same standards as journals and funders:

- NEJM AI editorial policy
- Oxford Bioinformatics author guidelines (Section 6.1)
- NIH grant application policy (NOT-OD-25-132)

Opening Poll

Raise your hand if:

1. You have used ChatGPT, Claude, or Gemini at least once
2. You have used AI to help write or debug code
3. You have used AI for anything related to research or coursework
4. You feel confident you could explain *how* an LLM works

No judgment — just getting a baseline.

Part 1

What are large language models, how do they work, and when should you use them?

What Is a Language Model?

At the most basic level, a language model answers one question:

> **Given everything written so far, what word (token) comes next?**

Repeat this billions of times with enough data and compute, and something remarkable emerges — a system that can reason, code, summarize, translate, and explain.

Key insight: LLMs do not *understand* language the way humans do. They produce statistically plausible continuations of text. This is both their power and their fundamental limitation.

📹

Recommended: 3Blue1Brown — But what is a GPT? Visual intro to transformers | Chapter 5, Deep Learning (~27 min)

A Brief History — Foundations

Year	Development	Reference
Pre-2010	n-gram models — count-based, very limited context	—
2013	Word2Vec — words as dense vectors	Mikolov et al., 2013
2014–2017	RNNs, LSTMs — sequential processing, longer context	Hochreiter & Schmidhuber, 1997
2017	**Transformer — attention mechanism; changed everything**	**Vaswani et al., NeurIPS 2017**

The transformer paper is the single most consequential development in modern AI.

A Brief History — The Modern Era

Year	Development	Reference
2018–20	BERT, GPT-1/2 — large-scale pre-training	Devlin et al., 2019
Nov 2022	**ChatGPT** — conversational AI for everyone	OpenAI, Nov 30, 2022
Mar 2023	GPT-4 — multimodal, major capability leap	GPT-4 Tech. Report, 2023
2023–25	Claude 3/4, Gemini, Llama 3 — diverse ecosystem	—
Nov 2025	Google Antigravity — agentic IDE, free preview	—
2026	Gemini 3.1 Pro · GPT-5.5 Instant · Claude Opus 4.7	—

Tokens and Context Windows

Tokens — the basic unit an LLM processes (~¾ of a word on average).

"biostatistics" ≈ `bio` + `stat` + `istics`   *"p-value"* ≈ `p` + `-` + `value`

Context window = total text the model can "see" at once.

Model	Context window
GPT-5.5 (default ChatGPT, May 2026)	1,000,000 tokens
Claude Opus 4.7	200,000 tokens (~150,000 words)
Gemini 3.1 Pro	1,000,000 tokens

Larger context → more history, longer code, bigger datasets to reason over.

How LLMs Are Trained

Step 1 — Pre-training: Read vast text (books, web, code, papers). Predict next tokens. Billions of parameters adjusted on trillions of tokens.

Step 2 — Supervised Fine-Tuning: Train on curated examples of good instruction-following.

Step 3 — RLHF *(Reinforcement Learning from Human Feedback)*
Human raters compare output pairs; model reinforced for preferred responses.

Christiano et al., NeurIPS 2017 · Ouyang et al., NeurIPS 2022

Result: A fluent, helpful assistant — that can still confidently produce incorrect information.

Major Models in 2025–2026

Organization	Model	Strengths
OpenAI	GPT-5.5 Instant, GPT-5.5	Default ChatGPT (May 2026); reduced hallucination on medical/legal/financial prompts
Anthropic	Claude Sonnet 4.6 / Opus 4.7	Long context; precise reasoning; strongest agentic coding
Google	Gemini 3 Pro / Gemini 3.1 Pro	1M token context; multimodal reasoning; Google ecosystem
Meta	Llama 3.1 / 3.3	Open-source; local deployment possible
Mistral	Mistral Large	Open-weight European alternative

USC students have free GPT-5.5 Instant access via the USC–OpenAI collaboration.

When to Use What Tool

Task	Approach
Write / debug R or Python code	LLM or agentic tool — highly effective
Explain a statistical concept	LLM — useful, but **always verify**
Fit a regression model	R/Python packages; AI assists only
Systematic literature review	AI-assisted screening; human for final inclusion
Causal inference, novel methods	Statistical expertise primary; AI as sounding board
IRB protocol or grant text	AI can draft; you verify, fill specifics, and disclose

> **Key question:** Is the AI generating *text* or *reasoning*? Text: helpful. Reasoning: verify.

Quick Check

Raise your hand if you agree:

> "An LLM is basically a very advanced search engine."

Hold your answer — we will discuss after.

What is the key difference between **retrieving** stored information and **generating** new text?

Part 2

What can it actually do? Where does it fall short?

How Biostatisticians Already Use AI (1/2)

Code writing and debugging

Generate R/Python/SAS code from plain language. Explain errors. Translate between languages.

Try: ChatGPT · Claude · Cursor · Google Antigravity · GitHub Copilot

Literature review and summarization

Summarize abstracts at scale, extract results from PDFs, surface related work.

Try: Semantic Scholar · NotebookLM · Perplexity · Consensus

How Biostatisticians Already Use AI (2/2)

Data wrangling

Write cleaning scripts from a description of the problem. Handle messy formats, missing values, and inconsistent coding — tasks that are tedious but not algorithmically novel.

Scientific writing

Draft methods sections, interpret results for lay audiences, translate findings for clinicians and policy makers. Specialized tools exist for academic and grant writing.

Try: Jenni.ai · Paperpal · Thesify · Grantable · Grail

Case Study 1: Code Assistance

What works well: Standard data manipulation, common tests (t-test, ANOVA), ggplot2 from natural language descriptions, translating code between R and Python.

What to watch for:

- Variable names may not match your actual data
- May use deprecated functions or packages
- Always run and inspect output — never copy blindly

Critical finding — sample size calculation:

Dobler et al. tested ChatGPT on the Schoenfeld formula across multiple runs.

Errors occurred in every single run. The output looked plausible — but was wrong.

Dobler et al., Stat Med 2025

Case Study 2: Rare Disease Diagnosis

ChatGPT reached or approached the USMLE passing threshold — but standardized exams are the easy case. *Kung et al., PLOS Digit Health 2023*

DeepRare — an LLM multi-agent system tested on **6,401 cases**, **2,919 rare diseases**:

Input to DeepRare	Top-1 diagnosis accuracy
Symptoms only (HPO terms)	57.2%
Symptoms + genetic data	70.6%

Zhao et al., Nature 2026

Beats Exomiser (53.2%) and the best general LLM (33.4%) — yet top-1 accuracy is still ~57–71%. **Verification stays essential.**

Case Study 3: AI-Assisted Literature Review

A JAMIA study evaluated LLM-based tools for evidence synthesis in health research:

JAMIA 2025, Vol. 32(6):1071

Key findings:

- AI tools substantially reduce manual screening burden in systematic reviews
- Performance varies by domain: high for common conditions, lower for rare or complex topics
- Human-in-the-loop remains essential — AI errors are not random and may be systematic

Bottom line: AI is a useful co-pilot for evidence synthesis, not a replacement for scientific judgment.

The Hallucination Problem

LLMs produce plausible-sounding content that can be factually wrong — and they do not flag it.

In biostatistics, this means:

- Invented citations with real-looking author names, journals, and DOIs
- Systematically wrong formulas applied with apparent confidence (e.g., Schoenfeld sample size errors in *every* trial)
- Results that vary across runs of the *same* prompt

> **Test in class:** Ask any LLM to cite 5 papers on a narrow biostatistics topic. Check if they exist. *Alkaissi & McFarlane, Cureus 2023 · Dobler et al., Stat Med 2025*

The Agentic Coding Landscape

Moving beyond chat — **agentic tools** plan, write, run, and iterate on code autonomously.

Tool	Access	Notes
Claude Code	Paid	Reads/writes files, runs terminal; most capable
Google Antigravity	Free preview	VS Code fork; Gemini 3.1; multi-agent view
Cursor / Windsurf	Free tier	VS Code forks; GPT-5.5 + Claude Opus 4.7
GitHub Copilot	Free (Edu)	Autocomplete + chat; not truly agentic

Lab uses browser-based tools; agentic IDEs are optional.

Think-Pair-Share (4 minutes)

Think individually (2 min), then discuss with a neighbor (2 min):

> Think about your current or planned research.
> Name **one task** where AI could genuinely help, and **one task** where you would be cautious — and explain why for each.

A few volunteers will share with the group.

Part 3

Journal policies, NIH rules, and the researcher's obligation

Why Ethics Matters Here

AI tools are powerful. That power creates new professional obligations.

- You can draft a methods section in 30 seconds. Are you responsible for its accuracy? **Yes.**
- AI can generate citations. Do you verify them? **Always.**
- Can you polish grant text with an LLM? NIH says: **substantially AI-generated applications are not acceptable.**

The researcher's obligation does not change because AI assisted you. Disclosure and verification are now **explicit requirements** of major journals and funders.

NEJM AI Editorial Policy

Source: ai.nejm.org/about/editorial-policies

- **Full disclosure required** — authors complete an online form (via Convey) published with the article, describing tools used and content generated
- **AI cannot be listed as an author**
- **AI-generated content cannot be cited as a primary source**
- **Authors are fully responsible** for accuracy and integrity of all AI-assisted content
- **Reviewers** must not upload manuscripts to AI services

Oxford Bioinformatics Policy

Source: academic.oup.com/bioinformatics/pages/author-guidelines §6.1

- **AI tools do not qualify as authors** — journal actively screens author lists
- AI use must be disclosed in the **cover letter** to the editor and in the **Methods or Acknowledgements** section
- **Reviewers are prohibited** from using LLMs to generate critiques or uploading manuscripts to any AI tool
- References the COPE position statement on authorship and AI

> Representative of a growing consensus across Oxford, Elsevier, Springer, and other major publishers.

NIH Policy: Grant Applications

NOT-OD-25-132 · *grants.nih.gov/grants/guide/notice-files/NOT-OD-25-132.html*

- Substantially AI-generated content in applications is **not acceptable**
- Limited use for administrative or formatting tasks is permitted
- Post-award detection → referral to the **Office of Research Integrity**; potential cost disallowance, grant termination
- NIH limits PIs to **6 new applications per year** — partly in response to AI-assisted mass submissions

NIH Policy: Peer Review

NOT-OD-23-149 · grants.nih.gov/grants/guide/notice-files/NOT-OD-23-149.html

Generative AI is **explicitly prohibited** in the NIH peer review process:

- Reviewers **cannot** use LLMs to formulate critiques
- Reviewers **cannot** upload grant content to any AI platform — confidentiality violation
- Applies to all study section reviewers, regardless of career stage

> The same rule applies in principle to journal peer review (Oxford, NEJM AI).
> Treat reviewer access to others' work as confidential — AI services are not.

The Stochastic Parrot Problem

LLMs produce statistically plausible text — they do not *know* anything. *Bender et al., FAccT 2021*

Risks for biostatistical research:

- **Hallucination** — invented citations, made-up statistics, nonexistent guidelines
- **Bias amplification** — training data reflects historical inequities in medical literature
- **Over-reliance** — fluent, confident text accepted without verification
- **Privacy leakage** — sensitive data submitted to commercial services

> **Fluency ≠ accuracy.**

A Framework for Responsible Use

Before: Check journal/funder policy · Turn off training data · Never submit confidential data

During: Verify every factual claim and every number · Keep records of prompts used

When publishing: Disclose tools and content generated · Accept full responsibility

Stage	Key action
Before	Policy check + privacy settings
During	Verify citations, formulas, and numerical results
Submitting	Disclose fully; your name = your responsibility

Eight guidelines for biostatisticians: Dobler et al., Stat Med 2025

Discussion

Scenario:

> A colleague uses ChatGPT to write the statistical methods section of a manuscript.
> They verify every sentence — it is accurate.
> They submit without disclosing AI use.

Is this acceptable?

- Raise your hand if **yes**
- Raise your hand if **no**
- Raise your hand if **it depends**

Part 4

How to ask well

Weak Prompts vs. Strong Prompts

Weak: *"Explain confounding."*

No context, no audience, no format. The AI makes all choices for you.

Strong:

> "You are a biostatistics professor. Explain confounding to a first-year MPH student who just learned regression, using an environmental health example. Structure your answer: (1) one-sentence definition, (2) concrete example, (3) how to address it, (4) one common mistake."

Same topic. Dramatically different result.

We will go deep on this in Week 2.

The RCTF Framework

Element	Question to ask yourself	Example
Role	Who should the AI act as?	"You are a biostatistics professor"
Context	What is the situation?	"I am writing a methods section for NEJM AI"
Task	What exactly do you need?	"Interpret this odds ratio for a clinical audience"
Format	How should the output look?	"Three bullet points, ≤2 sentences each"

Not every prompt needs all four — but adding them almost always helps.

Live RCTF demo + deeper practice next week (Week 2: Prompt Engineering).

Before Lab — Tuesday May 26

Please complete before arriving:

1. Sign in to **chatgpt.com** with your USC account (GPT-5.5 Instant access included)
2. Create a free account at **claude.ai**
3. Sign in to **gemini.google.com** with your Google account
4. Confirm **R and RStudio** are installed on your laptop
5. Watch **3Blue1Brown: "But what is a GPT?"** (~27 min) — youtube.com/@3blue1brown

We will walk through the privacy setup now — please follow along.

Privacy Setup: ChatGPT

Disable training data use:

1. Click your profile icon (top-right corner)
2. **Settings → Data Controls**
3. Turn **OFF** *"Improve the model for everyone"*

> Even with this off, OpenAI may retain conversations for safety review.
> **Do not submit** identifiable patient data, unpublished manuscripts, or confidential research to any commercial AI service. This applies equally to Claude, Gemini, and all browser-based tools.

Privacy Setup: Claude and Gemini

Claude.ai

- Anthropic does **not** use Claude.ai conversations to train models by default
- Confirm: Settings → Privacy
- Organizational accounts (Claude for Work) offer additional protections

Gemini

1. Go to **myactivity.google.com**
2. Find *"Gemini Apps Activity"*
3. Turn **off** activity saving

Personalize Your LLM

A brief context message makes responses far more relevant throughout the course.

ChatGPT: Settings → Personalization → "What would you like ChatGPT to know?"

Example message:

> *I am a biostatistics graduate student at USC, comfortable with regression, probability, and R. Give precise, technical answers. Use R by default for code. Flag claims that may need verification.*

Claude: Paste this at the start of a new conversation or project.

Week 1 Recap

- LLMs predict text — they do not understand. **Fluency ≠ accuracy.**
- Generative AI is useful in biostatistics: code, writing, literature review.
- **Verification is not optional.** You are responsible for the output.
- Journal policies and NIH guidelines require explicit disclosure.
- Strong prompts use RCTF; small changes make a large difference.

Next: Lab Tue 5/26 · Week 2 Thu 5/28 (Prompt Engineering) · **HW1 due 5/28**

Questions?

Lu Zhang
Division of Biostatistics · USC Keck School of Medicine
luzhangstat.github.io

Office hours by appointment

Lab 1 Preview

Tuesday, May 26 · SSB-114 · 1:00–2:50 PM

Bring: Laptop with R/RStudio · browser tabs open for ChatGPT, Claude, Gemini

Activities:

1. Platform comparison — same prompt, different AI tools
2. Prompt refinement — from weak to strong, step by step
3. Vibe coding in R — AI-assisted statistical analysis

Worksheet and starter code: `week1/lab/`